

STUDY NOTES: ADVANCED DATA STRUCTURES & ALGORITHMS

Comprehensive Exam Preparation & Conceptual Review Guide

CHAPTER 4: BALANCED SEARCH TREES & GRAPH ALGORITHMS

Course: CS-301 Advanced Algorithms | Semester: Fall 2026 | Prepared by: Department of Computer Science

1. CORE CONCEPTS & DEFINITIONS

Definition 1 (AVL Tree): An AVL tree is a strictly self-balancing Binary Search Tree (BST) where the difference between heights of left and right subtrees (Balance Factor) for any node cannot exceed 1.

Definition 2 (Dynamic Programming): An algorithmic technique that solves complex problems by breaking them into overlapping subproblems and memoizing intermediate solutions to avoid redundant computation.

2. KEY FORMULAS & COMPLEXITY RULES

- Balance Factor Formula: $BF(Node) = Height(Left_Child) - Height(Right_Child)$, where $BF \in \{-1, 0, 1\}$.
- Tree Height Limit: $Height \leq 1.44 * \log_2(n + 2) - 0.328$.
- Master Theorem: $T(n) = aT(n/b) + f(n)$. If $f(n) = \Theta(n^{\log_b(a)})$, then $T(n) = \Theta(n^{\log_b(a)})$.
- Search, Insertion, Deletion Time Complexity: $O(\log n)$ worst-case.

3. ILLUSTRATIVE EXAMPLES & TREE ROTATIONS

Case 1 (Left-Left Heavy): Solved with single Right Rotation around the pivot node.

Case 2 (Left-Right Heavy): Solved with Left Rotation on left child followed by Right Rotation on parent node.

Example: Graph traversal via Breadth-First Search (BFS) uses a FIFO Queue ($O(V+E)$), whereas Depth-First Search (DFS) utilizes a LIFO Stack or recursion.

4. CHAPTER SUMMARY

Self-balancing binary trees ensure worst-case $O(\log n)$ lookup operations, circumventing degenerated $O(n)$ linked-list degradation. Graph shortest-path problems are solved using Dijkstra's algorithm with priority queues in $O((V + E) \log V)$.

5. REVIEW QUESTIONS

1. Given an AVL tree node with balance factor +2 and child balance factor -1, what composite rotation is required?
2. Under what condition does Dijkstra's algorithm fail, and what algorithm should be utilized instead? (Hint: Negative edge weights -> Bellman-Ford).